NATE STEMEN

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Summary

Research engineer experienced in quantum error mitigation, circuit compilation, and translating quantum research into practical tools for current and near-term devices. Lead developer of `mitiq`, a widely-used open-source Python library for error mitigation.

Education

University of Waterloo MMath in Applied Mathematics — 2020–2022

- Thesis: *Quantum Circuit Compilation from the Ground Up* advised by Joel Wallman

New York University B.Sc. in Mathematics and Physics — 2013–2017

- Thesis: *An Investigation of Q-Balls* advised by Luciano Medina

Employment

Senior Software Engineer IonQ — *May* 2026–

- Responsible for internal and external error mitigation tooling.

Visiting Researcher QuSoft / University of Amsterdam — *Jan–March* 2026

- Performed computational search for efficent unitary decompositions of bottleneck of QFT on symmetric group.
- Prepped for and made release of first major release of `mitiq` 1.0.0.

Member of Technical Staff Unitary Foundation — *Mar* 2022–*Dec* 2025

- Technical lead and maintainer of the open-source **Python** library `mitiq` for quantum error mitigation (350k+ downloads, 170+ citations), responsible for core architecture, API design, 20+ releases, and review and integration of contributions from 90+ contributors.
- Designed and implemented a modular, two-step application API for error-mitigation techniques, informed by user interviews, making mitigation circuits and experimental overhead explicit and inspectable.
- Delivered 10+ talks and tutorials at major conferences and research institutions (PyData, SciPy, IEEE QCE, QuSoft), accelerating adoption of mitiq by quantum software engineers and researchers.

Software Developer Overleaf — 2017–2021

- Improved LaTeX autocomplete using statistical analysis of 40,000 open-source documents, enhancing user experience for 300,000+ daily users.
- Monitored and supported data migration from PostgresQL to MongoDB.

Summer Researcher New York University — 2016

- Used **Python** to numerically solve nonlinear Schrödinger equations modeling electromagnetic pulse propagation in nonlinear media.

Summer Researcher Yale University (PROSPECT Experiment) — 2014 & 2015

- Built an optical simulation in **C++** to optimize detector design and study light collection and uniformity.
- Implemented pulse-shape discrimination techniques in **Python** to improve neutrino event selection.

Publications

LaRose, R. et al. (Aug. 2022). Mitiq: A software package for error mitigation on noisy quantum computers. *Quantum* 6, p. 774. URL: https://doi.org/10.22331/q-2022-08-11-774.

McDonough, B. et al. (2022). "Automated quantum error mitigation based on probabilistic error reduction". In: *2022 IEEE/ACM Third International Workshop on Quantum Computing Software (QCS)*, pp. 83–93. arXiv: 2210.08611 [quant-ph].

Ashenfelter, J. et al. (2016). Background Radiation Measurements at High Power Research Reactors. *Nucl. Instrum. Meth.* A806, pp. 401–419. arXiv: 1506.03547 [physics.ins-det].

Ashenfelter, J. et al. (2015). Light Collection an Pulse-Shape Discrimination in Elongated Scintillator Cells for the PROSPECT Reactor Antineutrino Experiment. *JINST* 10.11, P11004. arXiv: 1508.06575 [physics.ins-det].

Service

Graduate Student Mentor University of Amsterdam — 2025 & 2026
Graduate Student Mentor University of Washington — 2024 & 2025
IEEE QCE 2025 Workshop organizer Quantum Software 2.1 — 2025
WERQSHOP Chair Organizer https://werq.shop — 2025
Quantum Computing Devroom Chair FOSDEM — 2025
SciPy 2025 Reviewer — 2025
QED-C mentor — 2023–2024
Equity, Diversity and Inclusion Committee University of Waterloo; IQC — 2021–2022
Strategic Plan Implementation Working Group University of Waterloo — 2021

Continuing Education

Advanced Representation Theory and Applications University of Amsterdam (audit) — *Feb–Mar* 2026
Advanced Quantum Algorithms University of Amsterdam (audit) — *Feb–Mar* 2026
Hands-on quantum error correction with Google Quantum AI Coursera — *Feb* 2026
CSE 599C: Quantum Learning Theory University of Washington (audit) — *Jan–Mar* 2025
CSE 534: Quantum info. and computation University of Washington (audit) — *Sep–Dec* 2024
Quantum Machine Learning Workshop QSciTech-QuantumBC — *Jan–Feb* 2022
Presenting Data and Information Edward Tufte — *Nov* 2019

Tools

Languages
- Python, JavaScript, SQL, Ruby, bash

Software
- git/GitHub, docker, Linux, MacOS, LaTeX

Quantum
- SDKs: Cirq, Qiskit, pyQuil, Qibo